

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000828		
	Version:	7.0; Most-Recent; Effective; CURRENT		
	Status:	Effective	Purpose:	Uncontrolled Copy
Class:	Control del Diseño		Effective date:	28-Apr-2016
Title:	SM DG57. Probe replacement			

EQUIPMENT

1. A new probe code 232214 for instruments **with** impact detector hardware. (See procedure 1).
A new probe code 231932 for instruments **without** impact detector hardware. (See procedure 2).
2. Pliers.

SET UP

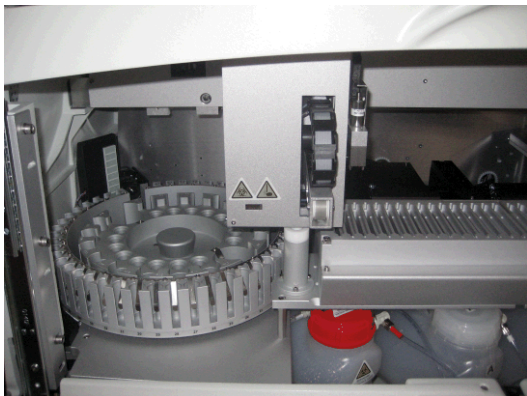
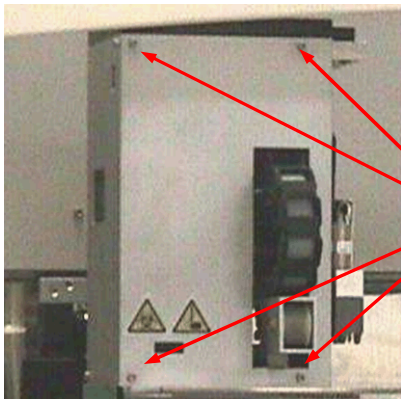
1. **Turn off** the instrument with the door opened.



WARNING!

There may be rests of blood in the Probe. Perform decontamination procedure REGD-0000787 before changing the probe.

PROCEDURE 1: Instrument with Impact Detector.

Process	Expected Results
1. Move the arm to the position showed in the Picture 1, in this position is possible to move the probe to its lower position without any interference with the elements of the instrument.  Picture 1 Unscrew and remove the four screws and remove the frontal cover of the robot.  Picture 2	The probe can be removed.

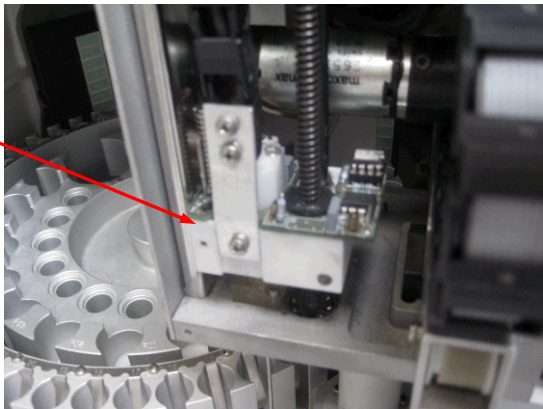
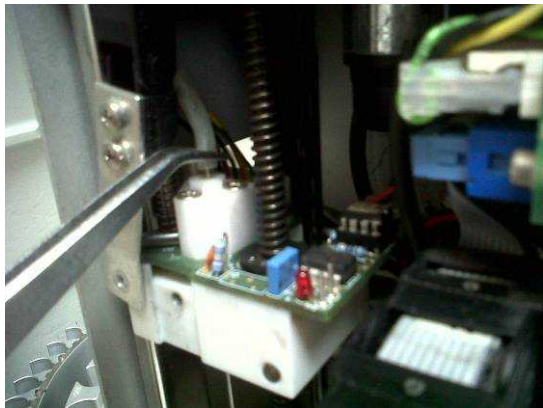
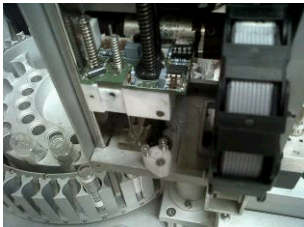
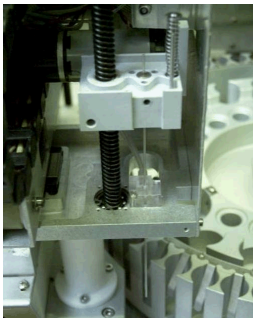
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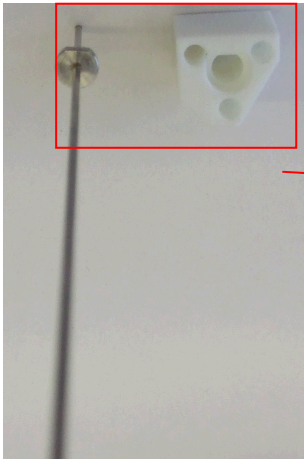
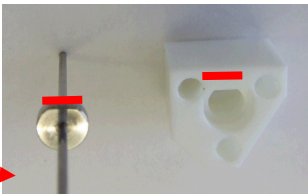
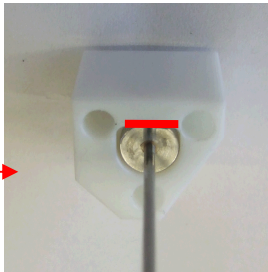
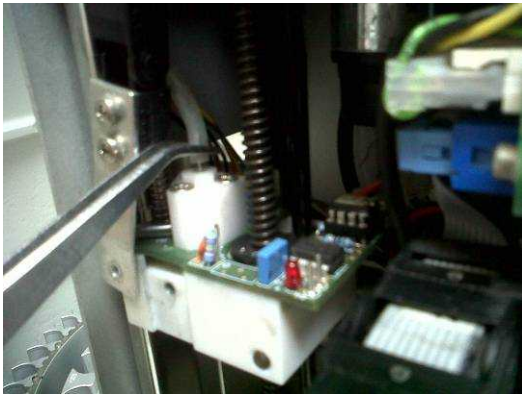
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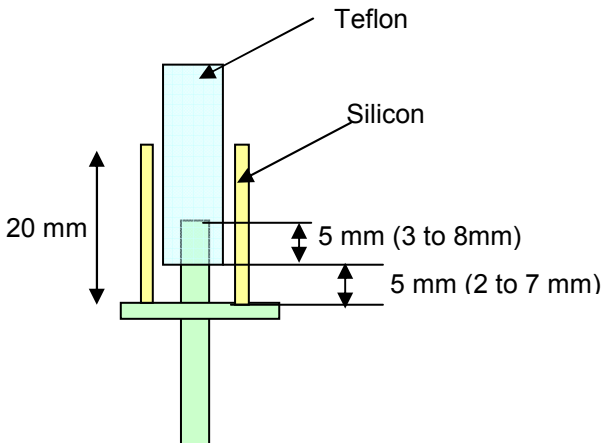
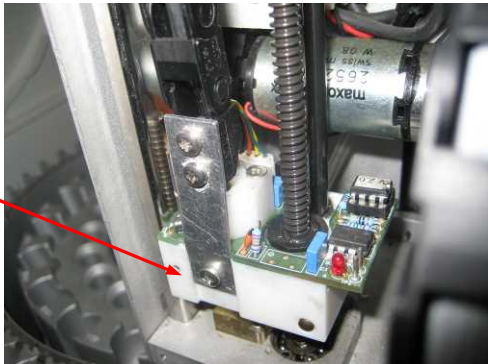
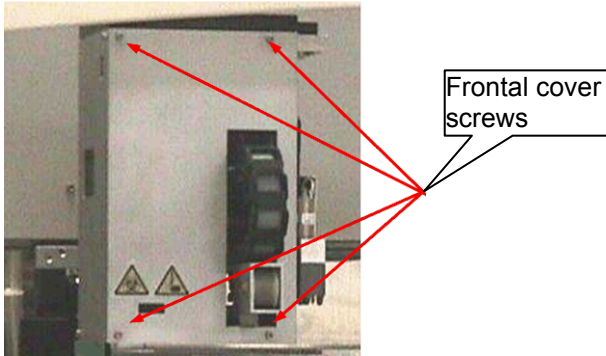
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Process	Expected Results
2. Unscrew and remove the chain bolt (the lower one that fixes it to the block).  Picture 3	---
3. Remove the Teflon tube from the probe and the connector, so that the chain is free.  Picture 4	The chain is free.
4. Unscrew and remove the three screws that are fixing the spring plastic support wear and the PCB and remove also the PCB.    Picture 5 Picture 6 Picture 7	Probe can be removed.
5. Remove the probe by pulling it upwards, replace by new one.	

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Process	Expected Results
6. Probe position can be placed only in one position due that spring plastic support and probe has a flat side in its shape.    Picture 8 Picture 9 Picture 10	---
7. Put the PCB in its place and fix it.  Picture 11	---
8. Connect the Teflon tube and the connector.  Picture 12	---

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Process	Expected Results
Needle's edge has to be inside Teflon tube (Picture 13) for approximately 5 mm (minimum 3 mm, maximum 8 mm). Check that Teflon tube has neither cuts nor shavings and that Silicon tube touches (± 2 mm) probe disc.  Picture 13	
9. Fix the chain with the bolt.  Picture 14	---
10. Place the frontal cover and fix it.  Picture 15	---

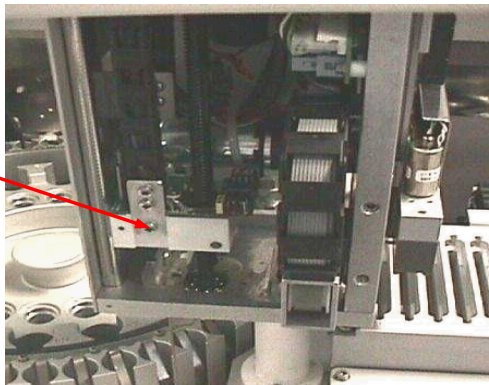
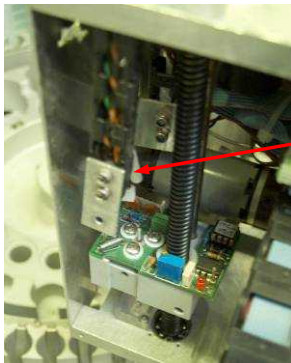
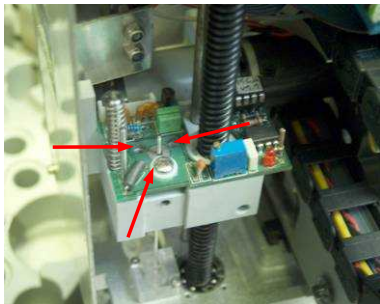
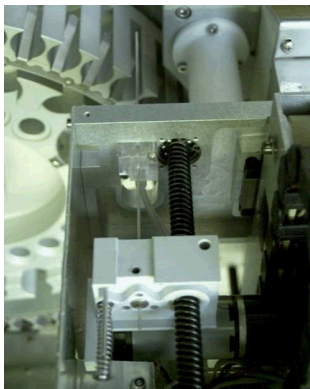
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Process	Expected Results
11. Follow procedure REGD-0000783 to adjust the parameters of probe and procedures REGD-0000807 and REGD-0001221 to adjust the probe positioning.	---
12. Optional step: In order to verify that the impact detector system has been properly assembled, Technical Service software can be used in one of the following ways: a) Using the 'Test' option in the 'Impact detector' section of the 'Handler' tab (see REGD-0000844) b) Using the 'Pipetting test' in 'Procedures' tab (see REGD-0000844) to aspire from a diluent bottle where the cap has not been removed. The front cover of the robot should be removed in order to have visual contact with the impact detector pca's LED during the test.	a) The test counter adds 1 count per provoked impact b) The red LED flashes each time an impact is provoked

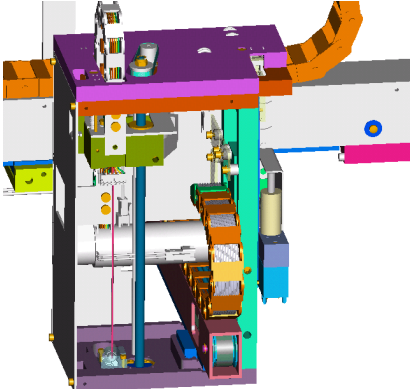
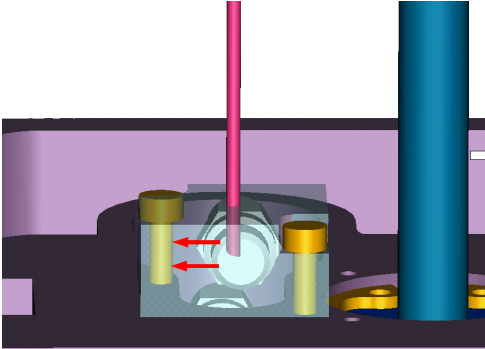
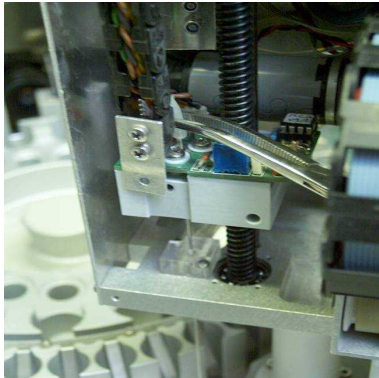
PROCEDURE 2: Instrument without Impact Detector.

Process	Expected Results
13. Move the arm to the position showed in the Picture 1, in this position is possible to move the probe to its lower position without any interference with the elements of the instrument. Picture 1 Unscrew and remove the four screws and remove the frontal cover of the robot. Picture 2	The probe can be removed.

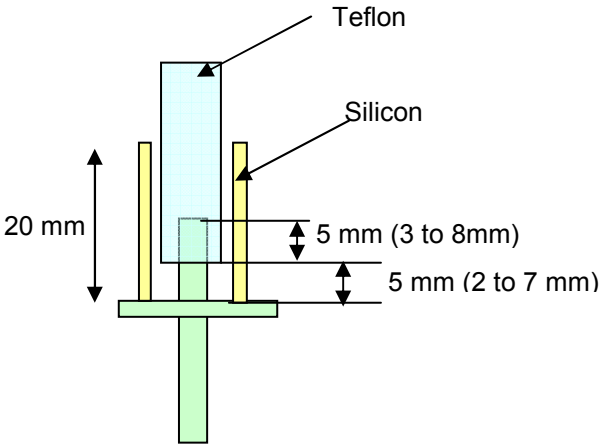
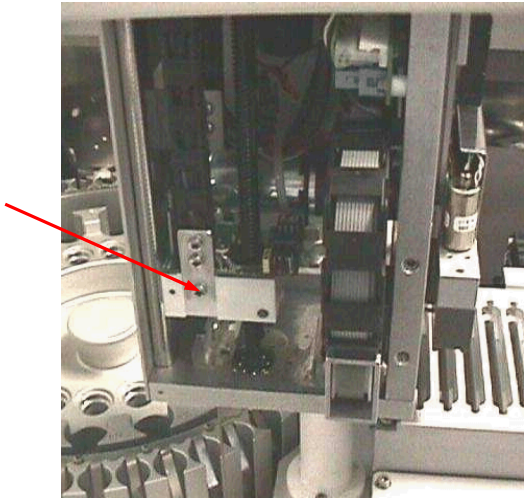
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Process	Expected Results
14. Unscrew and remove the chain bolt (the lower one that fixes it to the block).  Picture 3	---
15. Remove the Teflon tube from the probe and the connector, so that the chain is free.  Picture 4	The chain is free.
16. Unscrew and remove the three screws that are fixing the PCB and remove also the PCB.   Picture 5 Picture 6	Probe can be removed.
17. Remove the probe by pulling it upwards, replace by new one.	

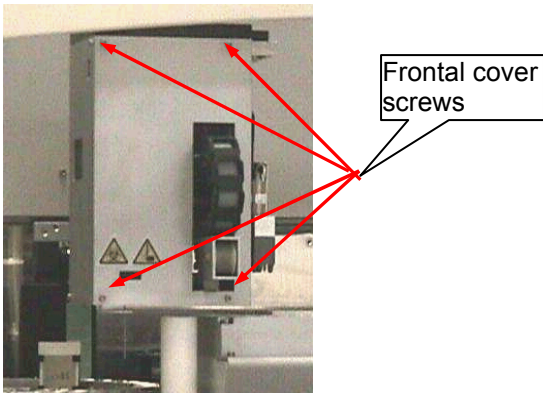
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Process	Expected Results
18. The orientation of the probe must be with the cut at the end of the probe orientated opposite to the lead screw.   Picture 7 Picture 8	---
19. Put the PCB in its place and fix it.  Picture 9	---
20. Connect the Teflon tube and the connector.   Picture 10	---

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Process	Expected Results
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21. Fix the chain with the bolt.  Picture 12	---

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Process	Expected Results
22. Place the frontal cover and fix it.  Picture 13	---
23. Follow procedure REGD-0000783 to adjust the parameters of probe and procedures REGD-0000807 and REGD-0001221 to adjust the probe positioning.	---

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DOCUMENT CHANGE HISTORY

Document change history			
Version	Date	Change	Rationale
3.0	2005/05/09	- REGD-0000828_v3 replaces document 57055V0M000 - Added references to probe positioning adjust procedures.	Uploaded document to GxPharma.
4.0	2011/04/28	Added procedure 1 for instruments with Impact Detector installed.	Updated the document due to the new impact detector hardware.
4.0	2011/04/28	Updated the Document Change History: - Date format changed to YYYY/MM/DD - Dates changed to document version effective date.	Adoption of ISO 8601 date format to avoid ambiguity.
4.0	2011/04/28	ID-GELSTATION and PROVUE erased from the title.	Service manual intended for Grifols market.
5.0	July 2011	Document REGD-0000828_v5 replaces document REGD-0000828_v4.	Document version changed.
5.0	July 2011	Step 6 inside procedure 1 has been modified.	Clarify explanation.
5.0	July 2011	Step 8 (procedure 1) and 19 (procedure 2) has been modified.	Ensure good connection between tube and needle.
5.0	July 2011	Document title.	Commercial reference substituted by project reference.
5.0	July 2011	Document header.	Increase information to ease document search.
6.0	December 2014	Added step 12: Impact detector verification	Verify the proper impact detector assembly after probe replacement
7.0	April 2016	Document title	Reference to DG-517 is deleted

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